



Department of Information Technology
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CD3291 & Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III / Rehashing
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** Jigsaw
- **Justification:**

Rehashing is one of the collision resolution techniques in hash data structure. Different types of hashing resolution techniques were already discussed in the class. Different problems were assigned to different group of students. The objective of JIGSAW activity is to promote both self and peer teaching which requires students to understand the concept very clearly and engage in discussion and learning.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - ✓ After I completed the collision resolution techniques, the students were given a group activity.
 - ✓ To accomplish this group activity the students were divided into 8 groups (Team A, Team B, Team C, Team D, Team E, Team F, Team G and Team H) with 8 members in each group as shown in fig 1 & 2.
 - ✓ One student from each group was selected and we framed the Expert Group and I delivered the instruction to the expert group students about how to perform rehashing and avoid collision.
 - ✓ The expert group members then discussed and instructed their original team members about the rehashing technique.
 - ✓ Each team was assigned with a question, they solved that question with the help of instruction delivered by expert member in their group and now all the members learned the concept and understand the way to handle collision by rehashing. This enabled the students to get insight knowledge in this topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO3	3	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
	3	2	1	2	2	2
Justification for correlation	Requires basic engineering knowledge to implement hash data structure	Able to identify engineering problems with hash data structures	Able to design solutions for complex engineering problems using hash data structure	Able to solve problem individually or in team and using hash data structures	Able to explain the way to solve the problem using hash table with their team members.	Able to apply hash data structures to solve some computational problem.

- **Images / Screenshot of the practice:**



Figure 1: Team A Students discussion with their team members [Dinesh Kumar, Varun, Sesu raj, Rasith Meeran, Hari Krishnan).



Figure 2: Team C team member Manish Vardhan explains the solution of rehashing problem to other students in the class

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students felt the activity is very interesting.
 - Students felt the activity conducted gave them a chance to improve their communication skills, also working as a team they can face group discussions in future work place.
 - Through learning from peer members, students felt comfortable to clarify their doubts.

- ❖ ***Benefit of the practice:***

- Students were actively participated in this activity.
 - Through this activity the students made to work as a team and explain to others also, like a group discussion activity.
 - From this activity, the students got clear idea about different hash resolution techniques

- ❖ ***Challenges faced in implementation:***

- Some students in the team did not participate actively in the team.
 - Individual assessment/observation was difficult.
 - It took some time and difficulties to organize the students into groups.

References:

- ❖ <https://www.jigsaw.org/>
 - ❖ <https://www.readingrockets.org/classroom/classroom-strategies/jigsaw>
 - ❖ <https://www.youtube.com/watch?v=euhtXUgBEts>
 - ❖ <https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/>

Signature of Faculty Member

HOD